[INCH-POUND]
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COMMERCIAL ITEM DESCRIPTION

MILLING MACHINE, VERTICAL KNEE TYPE, 3 AXIS, COMPUTER NUMERICAL CONTROL, 3.0 TO 7.5 HORSEPOWER SPINDLE DRIVE

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE

1.1 Scope. This commercial item description (CID) covers vertical knee type milling machines having computer numerical control of x, y, and z (quill) movement with equipment and accessories as described herein.

1.2 Intended use. The milling machines covered by this CID are intended for use in any metal cutting production shop where drilling, boring, tapping, reaming, and full (3-axis) contour milling operations are required.

2. CLASSIFICATION

2.1 Classification. The milling machine shall be of a vertical knee type having computer numerical control of x, y, and z (quill) axes. The machine shall be of the following classes, sizes, and grades. Class is determined by the spindle motor rating. Size is determined by the maximum table travel (X and Y axes) and quill travel (Z-axis). The class, size, and grade to be furnished shall be as specified (see 7.2).

Beneficial comments, recommendations, additions, deletions, clarification, etc. and any data which may improve this document should be sent to: Defense Supply Center Richmond, ATTN: DSCR-VBD, 8000 Jefferson Davis Highway, Richmond, VA 23297-5610.

AMSC N/A FSC 3417

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2.2 Class and size.

Class A - 3.0 horse power spindle motor rating

Size	X-axis travel, min.	Y-axis travel, min.	Z-axis travel, min.
1	24 in.	12 in.	5 in.
2	28 in.	12 in.	5 in.
3	30 in.	14 in.	5 in.

Class B - 5.0 horse power spindle motor rating

Size	X-axis travel, min.	Y-axis travel, min.	Z-axis travel, min.
1	31 in.	14 in.	5 in.
2	36 in.	14 in.	5 in.

Class C - 7.5 horse power spindle motor rating

Size	X-axis travel, min.	Y-axis travel, min.	Z-axis travel, min.
1	24 in.	14 in.	6 in.
2	30 in.	14 in.	6 in.

2.3 Grade. Grade A - High performance (see table I).
Grade B - Standard performance (see table I).

3. SALIENT CHARACTERISTICS

3.1 General requirements. The machine shall be designed and manufactured in accordance with the American National Standards Institute (ANSI) B11.8. The machine shall be new and one of the manufacturer's current models capable of operation in accordance with the requirements herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair. The machine configuration shall be consistent with current industry standards for a vertical knee type milling machine containing a column and base, knee, saddle, table, spindle, electric motor drive, and a computer numerical control (CNC) system. When specified (see 7.2), the machine shall be of a specific configuration (fixed head and column, swivel head, ram, or turret). The control system shall be a continuous path, closed-loop system providing full contouring capabilities on the machine's three linear axes. The term "full contouring", as used herein, shall be defined as a closed loop control system having simultaneous control of position, direction, and rate of movement of the machine's three linear axes in linear, parabolic, and circular interpolation. The machine shall provide a means for overload protection. The machine shall automatically stop spindle rotation and the motion of all slides upon an overload condition or spindle stall. The machine shall have a means to permit leveling and mounting to the floor or foundation. The axes of the machine shall be identified in accordance with Electronic Industries Association (EIA) 267.

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3.2 Measurement system. The U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the machine. When only one system of measurement is acceptable, the particular system required shall be as specified (see 7.2).

3.2.1 Measuring and indicating devices. The U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 7.2). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system. When specified (see 7.2), measuring and indicating devices shall be graduated in both the US and SI system of measurements.

3.3 Table. Unless otherwise specified (see 7.2), the number, size, and configuration of worktable T-slots shall conform to ANSI B5.1. The maximum rapid traverse speed shall not be less than 175 in/min and the maximum feed rate shall not be less than 100 in/min. If required, the following items shall be provided as specified (see 7.2).

a. Rotary table.
b. Index table.
c. Work light.

3.4 Spindle. Unless otherwise specified (see 7.2), the minimum spindle speed range shall be from 100 to 3500 revolutions per minute, in both the clockwise and counterclockwise directions. The spindle taper size shall be in accordance with ANSI B5.18.

3.5 Knee. Unless otherwise specified (see 7.2), the knee shall have a range of travel of not less than 14 inches. The knee shall be provided with a means for preventing dust and chips from entering the ways.

3.6 Lubrication. Mechanisms requiring periodic manual lubrication shall be readily accessible for servicing. All oil holes, grease fittings, and filler caps shall be accessible. When provided, recirculating lubrication system shall have a replaceable, in-line filter capable of removing particles 10 microns in size. A means for determining lubricant level in the reservoir(s) shall be by sight gauge, warning light, or a similar device.

3.7 Coolant system. Unless otherwise specified (see 7.2), the machine shall have a flood type coolant system. A means shall be provided to direct and control the amount of coolant. A base or floor-mounted splash tray shall be provided to catch coolant overflow from the chip and coolant channels around the working surface of the table.

3.8 Computer numerical control (CNC) system. The machine shall have a fully automated CNC system. The CNC system shall operate as a continuous-path, closed-loop system providing full contouring capabilities. The processing speed and temporary memory storage

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shall be of sufficient capacity to prevent delaying of the next incoming command or interrupting the machine's functions. The CNC system shall provide on-line diagnostics for detecting programming errors and problems with the control system. The CNC system shall be furnished with a postprocessor for converting specified part program language (see 7.2) to the form and format required for input to the CNC system. The CNC system shall be in accordance with all safety guidelines and procedures as specified herein.

3.8.1 Hardware features. Unless otherwise specified (see 7.2), the CNC system shall have not less than the following hardware features:

a. Operator control panel.
b. High resolution 9 inch monochrome display.
c. 3.5 inch, 1.44 Mbyte floppy disk drive.
d. 100 Mbyte hard-drive.
e. EIA RS-232C serial port.

3.8.1.1 Operator control panel. Unless otherwise specified (see 7.2), the operator control panel shall have not less than the following control features:

a. Emergency stop.
b. Feed hold.
c. Cycle start.
d. Feed rate override.
e. Spindle speed override.
f. Axis jog (continuous and intermittent).
g. Single block read.
h. Dry run.
i. Mode select.

3.8.1.2 Data display unit. Unless otherwise specified (see 7.2),the data display unit shall display not less than the following information:

a. Block sequence number.
b. Active block of stored data.
c. Actual position of each axis.
d. Feed rate and override condition.
e. Control/system error messages.
f. Graphics simulation (4 views) - XY, XZ, YZ, and isometric.

3.8.2 Operational modes.

3.8.2.1 Automatic mode. The automatic mode shall set the CNC system for full automatic "hands off" program operation.

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3.8.2.2 Manual data input (MDI) mode. The MDI mode shall permit manual entry of alphabetical, numerical, and special characters necessary for control of all programmable machine functions in accordance with EIA 274. The data shall be immediately displayed on the data display screen to permit data verification. The MDI mode shall have full editing capabilities. A means shall be provided for saving and recalling a part program developed in the MDI mode.

3.8.2.3 Conversational mode. The conversational mode shall provide the operator the ability to develop part programs by following computer prompts. A means shall be provided for saving and recalling a part program developed in the conversational mode.

3.8.3 Axis and tool management. Unless otherwise specified (see 7.2), the CNC system shall have not less than the following axis and tool management features:

a. Tool length offsets.
b. Cutter compensation.
c. Fixture compensation.
d. Reversal error compensation.
e. Lead error compensation.

3.8.4 Program and control features. Unless otherwise specified (see 7.2), the CNC system shall have not less than the following program and control features:

a. Spindle - off/clockwise/counter clockwise.
b. Coolant - on/off.
c. Reference zero.
d. Sequence number search.
e. Switchable inch/metric conversion.
f. Cartesian/polar coordinates.
g. Scale factor.
h. Mirror image.
i. Absolute/incremental axis positioning.
j. Interpolation - linear, circular, and helical.
k. Canned cycles - drilling, boring, tapping, and bolt hole.
l. Part rotation.
m. Dwell.

3.8.5 Variable data block format. The preparatory “G codes” and miscellaneous functions “M codes” shall be in accordance with EIA 274.

3.8.6 Program safe limits. The control system shall have the capability to place boundaries in a program beyond which the tool will not travel.

3.8.7 Battery back-up. If required, the CNC system shall have a battery back-up with a storage capacity as specified (see 7.2).

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3.9 Electrical requirements. The milling machine's electrical system shall conform ANSI/NFPA-79. The machine shall operate from a single 230/460 volt, 3 phase, 60 hertz (Hz) circuit. Unless otherwise specified (see 7.2), the machine shall be initially wired for 230 volt operation. All motors shall be energy efficient and shall conform to NEMA MG-1. The CNC system shall operate on a 120 volt, single phase,60 Hz circuit.

3.10 Performance. The machine shall meet the performance requirements shown in table I for a Grade A or Grade B machine as specified, and shall be tested in accordance with 5.3.1 through 5.3.3.

TABLE I. Performance requirements.

Test name	Parameter	Performance category	
		Grade A	Grade B
Linear displacement[1]	X, Y, and Z directions	0.0005 in/40 in	0.0015 in/40 in
Bidirectional repeatability	X, Y, and Z directions	±0.0001 in	±0.0003 in
Volumetric performance[1]	Diagonal displacement	0.001 in/40 in	0.003 in/40 in
Contouring performance (up to 80% of max. speed)	X-Y plane, 360° X-Z plane, 190° Y-Z plane, 190°	0.0005 in	0.0015 in
Squareness error (cw, ccw)	X-Y, X-Z, and Y-Z axes	2 arc-sec	5 arc-sec
Scale mismatch (cw, ccw) (with 10-inch-dia. ball bar)	X-Y axis	0.00025 in	0.001 in
Periodic angular positioning accuracy	A-axis B-axis	2 arc-sec	10 arc-sec
Bidirectional angular positioning repeatability	A-axis B-axis	0.5 arc-sec	5 arc-sec
Spindle dynamics	Radial error–average	0.0001 in	0.0003 in
	Radial error–asynchronous	0.0002 in	0.0005 in

[1] The machine's performance shall not be greater than specified for each performance category. If the axis travel is less than 40 inches, the tolerance shall be decreased proportionately.

3.11 Standard equipment. The manufacturer shall furnish all standard equipment.

3.12 Optional equipment. Optional equipment shall be furnished as specified (see 7.2).

3.13 Safety and health requirements. The manufacturer shall ensure the machine and all equipment and accessories used on the machine shall be in compliance with ANSI B11.8, OSHA 1910.212, and OSHA 1910.219. If a conflict arises between the ANSI and OSHA standards, the OSHA standards shall apply.

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3.14 Nameplate. A nameplate shall be securely attached to the machine. Unless otherwise specified (see 7.2), the nameplate shall contain the following information:

a. Nomenclature.
b. Manufacturer's name.
c. Machine model designation.
d. Power input (volts, total amps, phase, frequency).
e. Short circuit/over-current rating.
f. Contract number or order number.
g. National stock number or plant equipment code.
h. Date of manufacture.

3.15 Lubrication plate or chart. When specified (see 7.2), a lubrication plate or chart shall be attached to the machine. The information provided on the plate or chart shall include:

a. Points of lubricant application.
b. Servicing interval.
c. Type of lubricant(s) with SAE number or lubricant identifier.

4. REGULATORY REQUIREMENTS

4.1 Recycled, recovered, or environmentally preferable materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Environmental protection. The machine shall meet all applicable Environmental Protection Agency restrictions in effect on the date of the contract. These regulations apply to the emission of materials hazardous to the environment or the operator's health and shall be met during the manufacturing, service, transportation, storage, and operation of the machine.

5. QUALITY ASSURANCE PROVISIONS

5.1 Product conformance. The product provided shall meet the salient characteristics of this commercial item description; conform to the manufacturer's drawings, specifications, standards, and quality assurance practices; and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

5.2 Examination. The machine shall be examined to determine compliance with all requirements specified in this CID.

5.3 Testing. Unless otherwise specified (see 7.2), the machine shall be tested in accordance with sections 5.3.1 through 5.3.3.

5.3.1 Test conditions. The machine shall be tested in accordance with the environmental conditions as specified in section 3 of ASME B5.54.

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5.3.2 Operational test. The machine and its CNC system shall be operated in accordance with the manufacturer's standard operating procedures for warm-up and run-off checks. The machine shall be operated at no load for not less than 3 hours. Proper operation of all controls and features shall be verified during the trial period.

5.3.3 Performance tests. The performance tests shall be conducted in accordance with Appendix B of ASME B5.54. If a rotary table is included with the milling machine, the periodic angular positioning accuracy and the bi-directional angular positioning repeatability tests shall be conducted in accordance with ASME B5.54. The test results will be compared to the performance levels listed in table I.

5.4 Acceptance. Unless otherwise specified (see 7.2), preliminary and final acceptance testing shall be conducted at the manufacturer's site. The results of the performance tests will be compared to the performance criteria listed in table I. The machine shall meet or exceed the requirements listed in table I for the performance category selected. Failure to meet any of the requirements contained herein shall be cause for rejection.

6. PACKAGING

6.1 Packaging requirements. Preservation, packing, and marking shall be as specified (see 7.2).

7. NOTES

7.1 Source of documents.

7.1.1 Industry standards. Copies of industry standards referenced in this commercial item description may be obtained from the following addresses:

American National Standards Institute (ANSI)

ANSI B11.8 Safety Requirements for Construction, Care, and Use of Drilling, Milling, and Boring Machines and Machine Tools

ANSI B5.1 T-Slots, Their Bolts, Nuts, and Tongues

ANSI B5.18 Spindle Noses and Tool Shanks for Milling Machines

Applications for copies should be sent to: American National Standards Institute, 11 West 42nd Street, New York, NY 10036.

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National Fire Protection Association (NFPA)

ANSI/NFPA-79 Electrical Standard for Industrial Machinery

Applications for copies should be sent to: American National Standards Institute, 11 West 42nd Street, New York, NY 10036.

American Society of Mechanical Engineers (ASME)

ASME B5.54 Methods for Performance Evaluation of Computer Numerically Controlled Machining Centers

Applications for copies should be sent to: American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.

Electronic Industries Association (EIA)

EIA 232 Interface Between Data Terminal Equipment and Data Communication Equipment Employing Serial Binary Data Exchange

EIA 267 Axis and Motion Nomenclature for Numerically Controlled Machines

EIA 274 Interchangeable Variable Block Data Format for Positioning and Contouring/Positioning Numerically Controlled Machines

Applications for copies should be sent to: Electronic Industries Association, Engineering Department, 2500 Wilson Boulevard, Arlington, VA 22201-3834.

National Electrical Manufacturers Association (NEMA)

NEMA MG-1 Motors and Generators

Applications for copies should be sent to: National Electrical Manufacturer's Association, 1300 North 17th Street, Suite 1847, Rossyln, VA 22209.

U.S. Department of Labor, Occupational Safety and Health Administration (OSHA)

OSHA 1910.212 General Industry, Safety and Health Standards (29 CFR 1910)

OSHA 1910.219 General Industry, Safety and Health Standards (29 CFR 1910)

Applications for copies should be sent to: U.S. Department of Labor, 200 Constitution Ave., Room 423, NW, Washington DC 20210.

7.2 Ordering data. Acquisition documents must specify the following:

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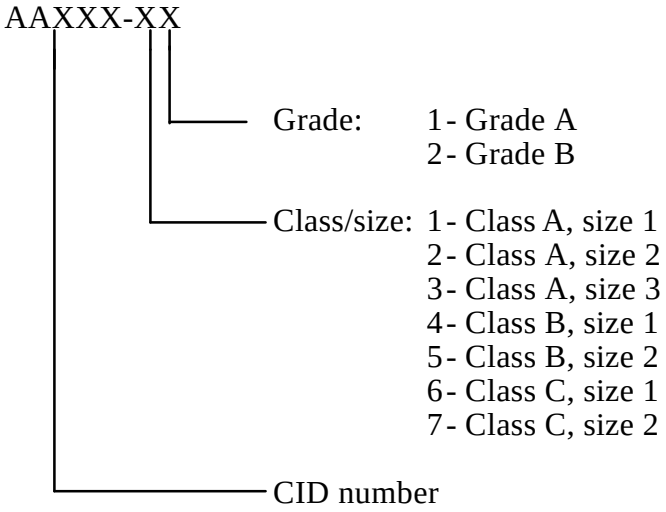
a. Title, number, and date of this document.
b. Class, size, and grade of machine required (see 2.1).
c. Machine configuration (fixed head and column, swivel head, ram, turret), if required (see 3.1).
d. Measurement system, if required (see 3.2).
e. Measurement system for indicating devices, if required (see 3.2.1).
f. Dual graduation of indicating devices, if required (see 3.2.1).
g. Work table T-slots, if different (see 3.3).
h. Rotary table, index table, and work light, if required (see 3.3).
i. Spindle speed, if different (see 3.4).
j. Knee travel, if different (see 3.5)
k. Coolant system, if different (see 3.7).
l. Part program language used, specify (see 3.8).
m. Hardware features, if different (see 3.8.1).
n. Operator control panel, if different (see 3.8.1.1).
o. Data display unit, if different (see 3.8.1.2).
p. Axis and tool management features, if different (see 3.8.3).
q. Program and control features, if different (see 3.8.4).
r. Battery back-up and storage capacity, if required (see 3.8.7).
s. Electrical requirements, if different (see 3.9).
t. Optional equipment, if required (see 3.12).
u. Nameplate, if different (see 3.14).
v. Lubrication chart, if required (see 3.15).
w. Performance testing, if different (see 5.3).
x. Acceptance criteria, if different (see 5.4).
y. Packaging, if different (see 6.1).

7.3 Part identification number (PIN). The following part identification numbering procedure is for government purposes and does not constitute a requirement for the contractor.



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Custodians
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Navy - SH

Reviewers
Air Force - 84

PREPARING ACTIVITY:
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